

Product Carbon Footprint Study Report

Product Name: keyboard

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: keyboard

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1Item keyboard

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1Item keyboard.

2. System Boundary

keyboard production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of keyboard produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **29.760 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

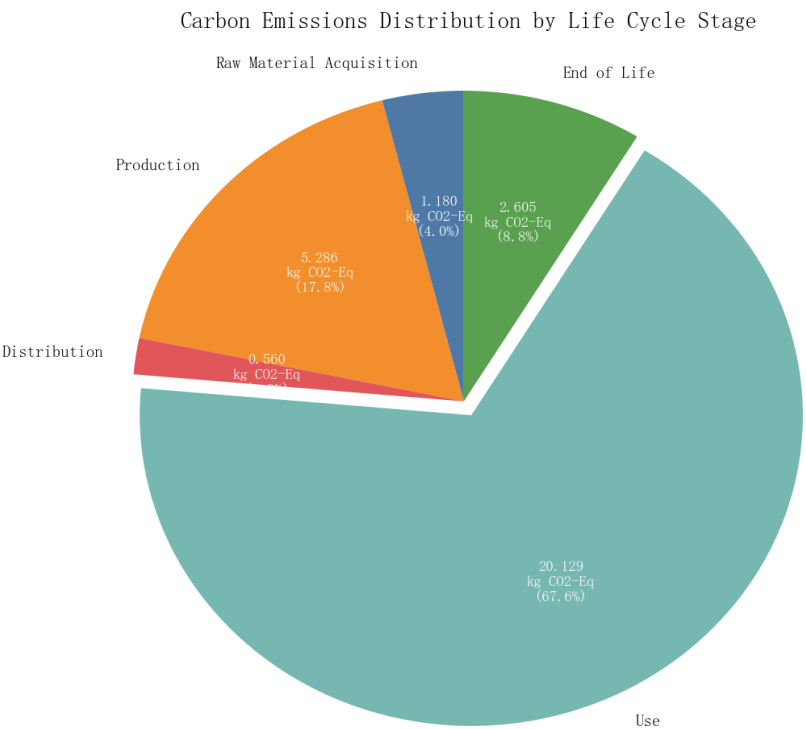


Table 1: Carbon Emissions by Life Cycle Stage

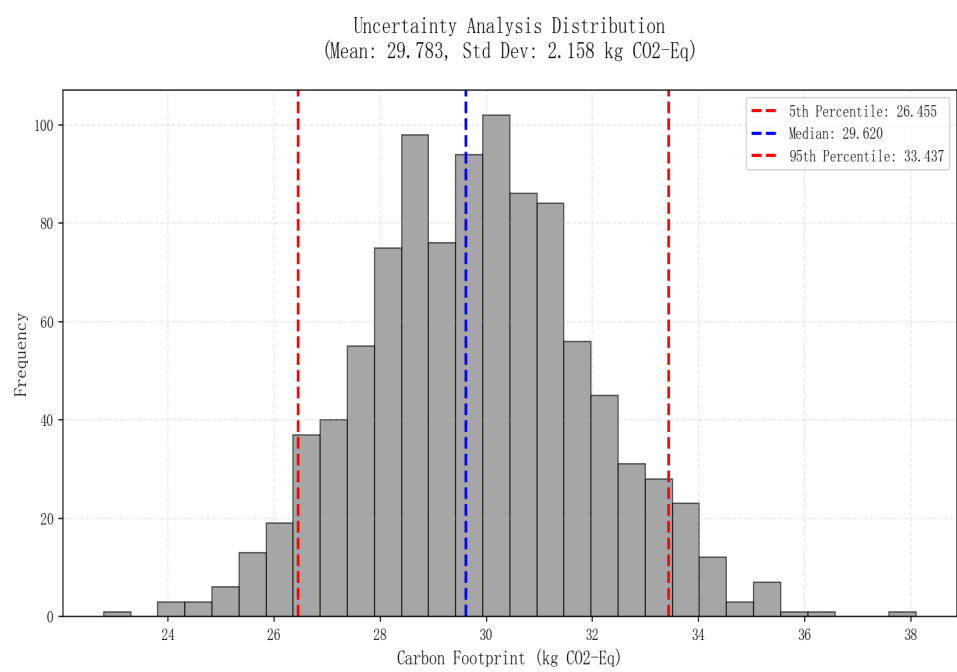
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.180	3.97
Production	5.286	17.76
Distribution	0.560	1.88

Use	20.129	67.64
End of Life	2.605	8.75
Total	29.760	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	29.783
Standard Deviation	2.158
Coefficient of Variation (CV)	7.24%
Median	29.620
5th Percentile	26.455
95th Percentile	33.437

VI. Interpretation of Results

1. Description of Results

The keyboard exhibits a total carbon footprint of 29.760 kg CO₂-Eq over its lifecycle, representing a moderate impact level for an electronic input device when benchmarked against similar consumer electronics. This value reflects a comprehensive cradle-to-grave assessment, with the use phase emerging as the dominant contributor at 20.129 kg CO₂-Eq (67.6%), followed by production at 5.286 kg CO₂-Eq (17.8%) and end-of-life at 2.605 kg CO₂-Eq (8.8%). Raw material extraction and distribution phases contribute minimally, accounting for only 4.0% and 1.9% respectively.

Lifecycle stage contributions highlight a use-phase intensive profile typical of electrically powered peripherals, where operational energy consumption drives the majority of emissions. The relatively low raw material and distribution shares indicate efficient upstream supply chains, while the end-of-life contribution suggests opportunities in waste management. Major emission sources are concentrated in the use stage, likely stemming from electricity demand during active and standby modes, underscoring the need to address energy efficiency in product design and user behavior.

2. Statement of Assumptions and Limitations

This study assumes a standard product lifespan of five years with average daily usage patterns of four hours, employing a representative European electricity grid mix for the use phase. Additional assumptions include generic material compositions for plastics and electronics, as well as standard transportation distances for distribution. End-of-life modeling presumes a mix of recycling, incineration, and landfilling consistent with EU waste directives.

Data sources primarily rely on secondary databases such as Ecoinvent and industry averages, introducing limitations in specificity for proprietary manufacturing processes and regional variations in material sourcing. Data quality is constrained by temporal gaps in emission factors and potential underrepresentation of emerging low-carbon technologies. System boundary restrictions encompass cradle-to-grave stages but exclude indirect upstream activities such as employee commuting and capital equipment depreciation, as well as downstream consumer behavior beyond direct product use.

3. Suggestions for Improvement

- Target the use phase by integrating low-power components and automatic sleep modes to reduce electricity consumption by up to 30%, directly lowering the 67.6% emission share.
- Shift to recycled plastics and bio-based materials in production to cut raw material and manufacturing impacts while maintaining structural integrity.
- Optimize end-of-life processes through design for disassembly and partnerships with certified recyclers to minimize the 8.8% contribution from waste treatment.
- Implement supplier-specific primary data collection for production and raw material stages to enhance accuracy beyond generic database values.
- Explore renewable energy sourcing for manufacturing facilities and user education campaigns on energy-efficient usage to achieve broader emission reductions.

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